

Potential Across Random Permutations

A companion note on when disorder budgets help search

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Abstract. The first paper bounds the cost of one search when an array’s rank-displacement potential is at most E ; the second considers repeated searches and a reusable index. Here we put that budget in context for a uniformly random permutation. Exact enumeration at $n = 9$ illustrates the distribution, while exact moments and a tail bound show that a typical uniform shuffle lies beyond the point where the potential-only displacement bound narrows the search window.

Potential under uniform shuffling

Let $\pi = (\pi_1, \dots, \pi_n)$ be a uniformly random permutation of the ranks $1, \dots, n$. The values of the underlying distinct keys do not matter. Their disorder potential is

$$V_n(\pi) = \frac{1}{2} \sum_{i=1}^n (\pi_i - i)^2 = \sum_{i=1}^n i^2 - \sum_{i=1}^n i\pi_i.$$

Since $\mathbb{E}\pi_i = (n+1)/2$, $\text{Var}(\pi_i) = (n^2 - 1)/12$, and $\text{Cov}(\pi_i, \pi_j) = -(n+1)/12$ for $i \neq j$,

$$\mu_n = \mathbb{E}V_n = \frac{n(n^2 - 1)}{12}, \quad \sigma_n^2 = \text{Var}(V_n) = \frac{n^2(n-1)(n+1)^2}{144}.$$

Complementing every rank gives an exact pairing:

$$V_n(\pi) + V_n(n+1-\pi) = \frac{n(n^2 - 1)}{6} = 2\mu_n.$$

Thus the finite distribution is symmetric about its mean. It need not be exactly normal: at $n = 9$ it has two modes. A combinatorial central limit theorem for permutation sums gives $(V_n - \mu_n)/\sigma_n \Rightarrow N(0, 1)$ as n grows. For a fixed tail probability δ , a normal approximation suggests $E_\delta \approx \mu_n + \sigma_n \Phi^{-1}(1 - \delta)$, where Φ is the standard normal distribution function. Exact counts are preferable in small samples and in the tails.

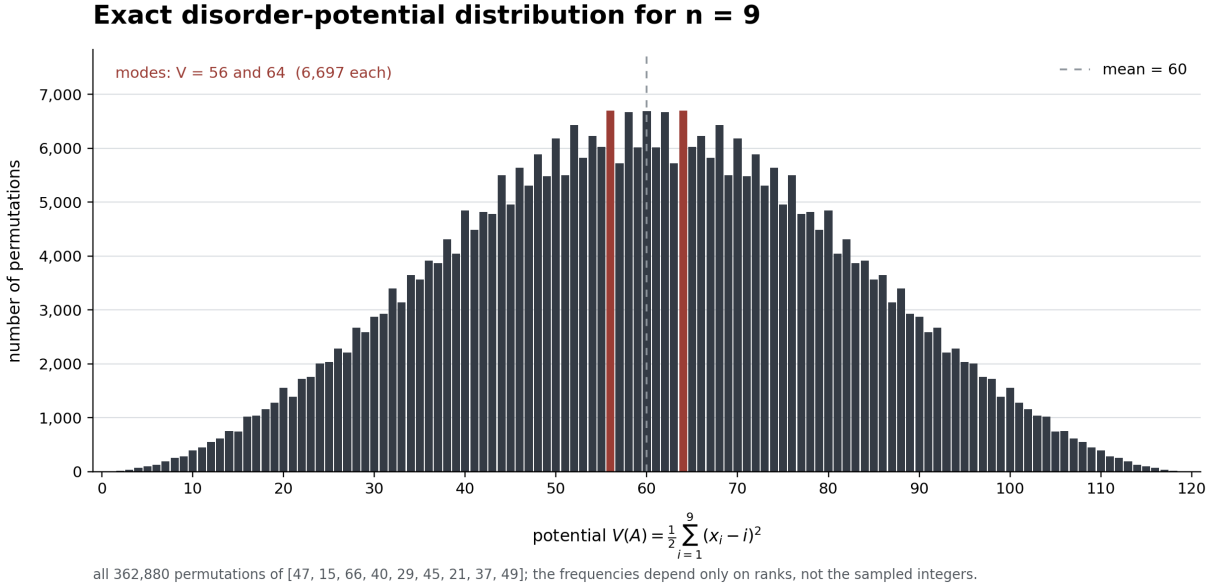


Figure 1: Exact counts for all $9! = 362,880$ orderings of nine distinct keys. Every integer potential from 0 to 120 occurs. The mean is 60; the modes are 56 and 64, each with 6,697 orderings.

What the distribution says about search

The first paper proves that a supplied promise $V(A) \leq E$ gives tight worst-case one-search cost

$$Q^*(n, E) = \Theta\left(\log n + \min\{n, \sqrt{E}\}\right).$$

The repeated-search note gives the upper bound

$$\begin{aligned} T_q^*(n, E) &= O(q \log n + \min\{qR_E, n \log(R_E + 1), n[1 + \log(1 + E/n)]\}), \\ R_E &= D_E + 1, \end{aligned}$$

where the displacement certificate implied by the promise is

$$D_E = \min \left\{ n - 1, \left\lfloor \frac{\sqrt{1 + 8E} - 1}{2} \right\rfloor \right\}.$$

The distribution of V_n tells us what fraction of uniform shuffles a chosen E covers. Coverage is not a certificate for an individual input: its promise must be supplied or verified, and verification comparisons must be counted.

The displacement certificate saturates at $D_E = n - 1$ when $E \geq E_{\text{sat}} = n(n - 1)/2$. Yet $\mu_n \sim n^3/12$ while $E_{\text{sat}} \sim n^2/2$. For $n > 5$, Chebyshev's inequality makes the asymptotic separation explicit:

$$\Pr(V_n < E_{\text{sat}}) \leq \frac{\sigma_n^2}{(\mu_n - E_{\text{sat}})^2} = \frac{(n + 1)^2}{(n - 1)(n - 5)^2} = O(1/n).$$

At $n = 9$, $E_{\text{sat}} = 36$ and the exact fraction below it is $48,913/362,880 \approx 13.5\%$. The low-disorder cases that receive a narrower window from this scalar certificate become rare under uniform shuffling. Search gains for typical inputs therefore call for a distribution favoring near-sorted arrays or a more informative description of where their disorder lies.

Reproduction. The experiment README, exact counts, and enumeration script accompany this note.